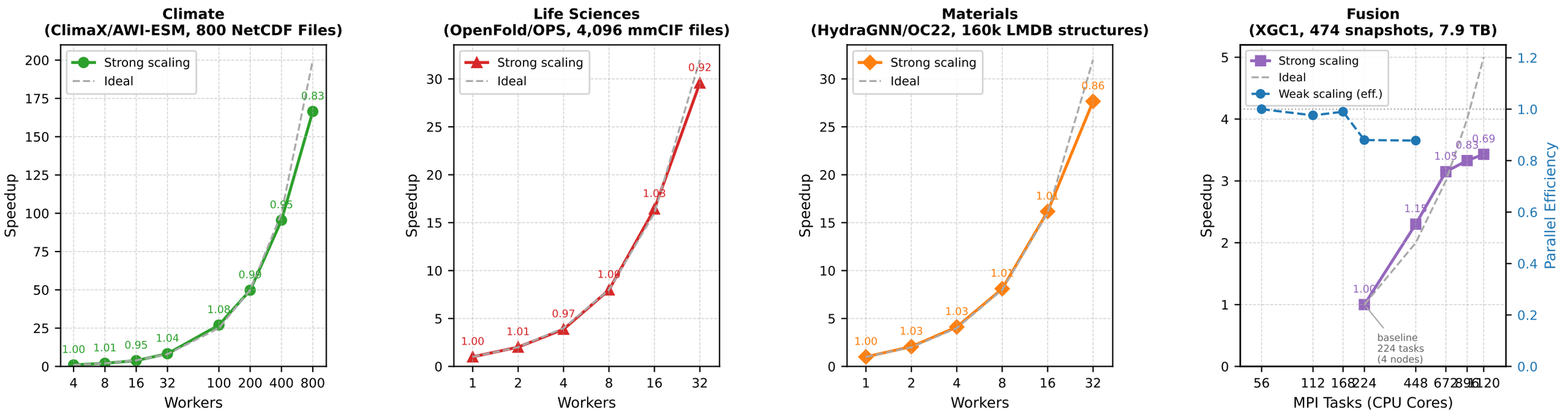
Climate
(ClimaX/AWI-ESM, 800 NetCDF Files)
Strong scaling
Ideal
Speedup
Workers
1.00
1.01
0.95
1.04
1.08
0.99
0.95
0.83
Life Sciences
(OpenFold/OPS, 4,096 mmCIF files)
Strong scaling
Ideal
Speedup
Workers
1.00
1.01
0.97
1.00
1.03
0.92
Materials
(HydraGNN/OC22, 160k LMDB structures)
Strong scaling
Ideal
Speedup
Workers
1.00
1.03
1.03
1.01
1.01
0.86
Fusion
(XGC1, 474 snapshots, 7.9 TB)
Strong scaling
Ideal
Weak scaling (eff.)
Speedup
Parallel Efficiency
MPI Tasks (CPU Cores)
1.00
1.15
1.05
0.83
0.69
baseline
224 tasks
(4 nodes)